Mst Afsana Mimi

Amherst, MA, USA | mmimi@umass.edu | +1 207 437 8137 | github.com/mstafsanamimi
linkedin.com/in/mst-afsana-mimi-578234b1/ | mstafsanamimi.github.io

Professional Summary

Master's graduate in Statistics from the University of Massachusetts Amherst with research and teaching experience in statistical analysis, data management, and quantitative reporting. Background includes academic research support, interdisciplinary collaboration, and use of Python, R, SQL, SAS, SPSS, and Stata for data analysis. Interested in data analyst roles that support evidence-based decision-making in higher education and research-focused environments.

Education

University of Massachusetts Amherst, Master's in Statistics (CGPA: 3.77) Sep 2023 – May 2026

- **Advisor:** Lulu Kang
- **Project: Gaussian Process Regression with Fairness Constraint** The project focuses on incorporating fairness constraints into Gaussian Process models that handle both qualitative and quantitative variables to reduce bias in predictions. It aims to ensure fair and reliable performance across protected groups while maintaining predictive accuracy.

University of Dhaka, Master's in Applied Statistics (CGPA: 3.75) Jan 2017 – Dec 2018

- **Advisor:** Md Hasinur Rahaman Khan
- **Project: Variable selection for censored data using modified correlation adjusted correlation (mcar) scores.** We proposed Modified Correlation Adjusted Correation (MCAR) scores for the variable selection procedure for censored high-dimensional data with AFT models. MCAR extends the CAR scores method by incorporating Huang and Frylewicz's NOVEL integration of sample and threshold correlation estimators. Its performance was validated through simulations and real microarray data (DLBCL), demonstrating improved accuracy over existing approaches.

University of Dhaka, Bachelor's in Applied Statistics (CGPA: 3.61) Jan 2013 – Dec 2016

Current Project

Can Agentic AI Eliminate the Need for a Dedicated Relational Foundation Model? Developing an agentic machine-learning framework for relational databases to automate schema understanding, relational evidence construction, and predictive reasoning over structured data. The project aims to evaluate relational prediction workflows and develop a future benchmark for testing agentic systems on realistic database tasks.

Experience

Teaching Assistant, Department of Mathematics and Statistics, University of Massachusetts Amherst– Massachusetts, USA. Sep 2023 – May 2026

- Support instruction in mathematics and statistics courses and help students build understanding of quantitative concepts and problem-solving methods.
- Communicate technical material clearly in academic settings, strengthening skills in explanation, organization, and detail-oriented review.

Research Assistant, Department of Computer Science and Engineering, University of Dhaka – Dhaka, Bangladesh. July 2021 – June 2022

Project: *Development of A Robust Bangla Language Model and a Bangla NLP Server (Funded by Centennial Research Grant, University of Dhaka)*.

- Contributed to an interdisciplinary research project involving data-focused work in collaboration with faculty and researchers in computer science.
- Supported research activities that required careful handling of project data, analytical thinking, and preparation of research-related outputs.

Research Assistant, Institute of Statistical Research and Training, University of Dhaka – Dhaka, Bangladesh. July 2018 – June 2019

Project: *Poverty and Inequality Mapping in Bangladesh.*

- Supported statistical research on poverty and inequality mapping, contributing to analysis workflows used in an applied social-impact context.
- Worked in a research environment that required data organization, quantitative analysis, and clear presentation of findings.

Technical Skills

Programming Languages: Python, R, C, SQL
Statistical Software: SAS, SPSS, Stata
Data Analysis: Statistical modeling, data management, quantitative research, reporting, data cleaning
Libraries/Tools: PyTorch, Torch Geometry

Awards and Scholarship

- **Conference Awards for Scientists (CASc)** by International Society for Clinical Biostatistics (ISCB) for "Variable selection for censored data using modified correlation adjusted correlation (mcar) scores."
  This award has been received at the 40th annual conference of International Society for Clinical Biostatistics (ISCB, 2019) in Leuven, Belgium in July 2019.
- **National Science and Technology (NST) Fellowship Award** by Education Board of Bangladesh for MS thesis.
- **Dhaka University Merit Scholarship** for academic results in B.S. Hon's examination.

Publications

- Mimi, A., & Khan, M. H. R. (2021). Variable selection for censored data using modified correlation adjusted correlation (mcar) scores. Statistics in Medicine, 40(23), 5046–5064.

Google Scholar: https://scholar.google.com/citations?user=LaVyjwYAAAAJ&hl=en&oi=sra